Dear Hiring Manager,

I am excited to apply for this position, bringing an Industrial Engineering background with hands-on experience in AI, machine learning, data science, generative AI, and geospatial analytics. My work spans research and professional environments, where I develop end-to-end AI and data-driven solutions using Python, statistical modeling, machine learning, large language models, and data visualization.

At Oak Ridge National Laboratory, I develop machine learning solutions for scientific and engineering applications, including fine-tuning Swin Transformers for weather-station quality screening, developing deep learning models for atmospheric and geospatial datasets, automating geospatial dashboard generation through natural-language knowledge-graph reasoning, and deploying retrieval-augmented generation systems across large scientific literature collections. I also work with spectral and spatial analysis for ToF-SIMS data and automate parameter selection for materials characterization.

My academic research at the University of Tennessee further strengthened my ability to apply machine learning, statistics, optimization, and AI to complex datasets. I have developed graph neural networks for transportation and energy forecasting, AI-powered digital twins, and real-time optimization systems. My experience also includes practical machine learning using clustering, regression, tree-based models, CNNs, Vision Transformers, and statistical modeling.

I also bring professional data engineering experience, including scalable ETL/ELT pipelines, distributed processing with Spark and Hadoop, cloud-based data platforms, data modeling, and analytics. Earlier roles involved developing Power BI and Tableau dashboards and translating data into actionable insights for business and operational decision-making.

My background aligns particularly well with the position's emphasis on generative AI, LLMs, agentic systems, statistics, Python-based machine learning, and geospatial applications. I have hands-on experience with RAG, LangChain, LangGraph, LiteLLM, vLLM, Ollama, QLoRA fine-tuning, PyTorch, TensorFlow, AWS, and geospatial technologies.

Beyond technical skills, I bring strong communication, presentation, problem-solving, and teamwork abilities. My research and professional work have required communicating technical results across engineering, scientific, and non-technical audiences. I would welcome the opportunity to bring this combination of AI engineering, data science, and applied problem-solving to your team.

Sincerely,

Jose Alberto Tupayachi Silva